



Relationship lobbying through repeated contributions: a humanomics approach

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Abstract

Previous scholarly investigations of the effectiveness of political lobbying are abundant but have not yet reached a consensus. This study incorporates the work of Adam Smith, Vernon Smith, and Deidre McCloskey to consider the question from a new perspective, that of humanomics, with its emphasis on the efficacy and significance of human relationships. In doing so, we test the proposition that lobbying is neither a one-time quid pro quo nor reducible to dollars and cents but most often is based on a relationship between the lobbyist and the lobbied that has developed over time. We explore the impact of a more powerful executive branch by analyzing the efforts to lobby people who are or may become governors of US states. We estimate the effect of new term limits for state legislators (adopted from 2000 to 2015) on political donations to governors, lieutenant governors, attorneys general, and secretaries of state. If, as we suspect, shorter terms of office for legislators undermine the likelihood of durable lobbyist-legislator relationships, it follows that lobbyists will shift their focus, proxied by repeat-contribution behavior, toward the executive branch. Our findings indicate that that, indeed, is what happens, offering empirical evidence that relationship-building over time is a key component of the effort to exercise political influence.

Keywords Social Capital · Lobbying · Rent seeking · Regulatory capture · Term limits · Relationship lobbying · Cognitive capture

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1 Introduction

If you're a lobbyist who never gave us money, I didn't talk to you. If you're a lobbyist who gave us money, I might talk to you. — Mick Mulvaney (Former US Congressman)

While lobbying in the United States is pervasive, de Figueiredo and Richter (2014) claim that the dollar amount, about \$10 billion at the federal level, is comparatively small relative to what one might expect given the attention it receives.¹ For example, according to a 2000 study, firms give about 100 times more money to charity than they spend on lobbying (Milys et al., 2000). Furthermore, 90% of firms that could lobby opt not to do so (de Figueiredo & Richter, 2014). Indeed, the extensive scholarly literature on the perennial topic of money in politics reveals conflicting results about its influence on policy outcomes.² Interestingly, Ansolabehere et al. (2003) find that research on political action committees (PACs) shows that contributions to them are ineffective or even detrimental. That is, higher campaign contributions may correlate with fewer aligned policy votes. In contrast, Grier et al. (2023) find that when contribution patterns change, roll call votes also tend to change. Specifically, roll call vote patterns differed in one case related to sugar policies that Grier et al. (2023) studied. De Figueiredo and Silverman (2006) report that campaign contributions have larger effects on a particular university's funding when the legislator representing the university's district is also a sitting member of the appropriations committee. Work by Baumgartner et al. (2011) suggests that lobbying and money matter only in certain cases, such as those relating to taxes or those for which outcomes are determined by close votes.³

Many researchers conclude that it is not the influence of lobbying but a legislator's party along with district-level fixed effects that accurately predict legislative votes. Those scholars argue that, at the federal level at least, a legislator's DW-NOMINATE score (Poole, 2007; Poole & Rosenthal, 1985) is the best vote predictor, not the identities of campaign contributors nor the sizes of their contributions.⁴ An argument favoring predictive value is that DW-NOMINATE scores measure more than partisanship but also inter-party ideological variation (McCarty, 2016). However, as is also true with several other factors commonly entered to explain policy outcomes, many studies report null results between DW-NOMINATE scores and legislative votes (Lowery, 2013). Moreover, Stratman (1992) finds that PACs donate the most money to members of Congress who are likely to be undecided on an upcoming vote and give less but still positive amounts to members representing districts or states with large farm populations. Interestingly, Austen-Smith (1987) argues that such members may receive more campaign contributions because accepting those

¹ Source: OpenSecrets.org. <https://www.opensecrets.org/news/2022/11/total-cost-of-2022-state-and-federal-elections-projected-to-exceed-16-7-billion/> (accessed: 7/13/2023). See also Tullock (1980).

² While most of the relevant research focuses on large corporate or union donors, small donors matter more than may be generally assumed. The findings suggest that small individual donors contribute not only the average dollar in politics, but the marginal dollar as well (Ansolabehere et al., 2003).

³ Not just lobbying legislators but other branches of government, including bureaucrats (Furlong, 1998; McKay, 2011).

⁴ The idea is that, at least empirically, most roll call votes can be explained by considering only two dimensions of a dynamic, weighted NOMINATE score. They are the traditional left–right dimension and some salient issue of the day. Poole and Rosenthal (2001) themselves point out that the first dimension is usually highly correlated with the second dimension.

contributions signals a legislator's commitment to voters. Calcagno and Jackson (1998) argue that PACs focus money on senators deciding whether to participate in roll call votes.

Traditional neoclassical economic studies of money in politics report inconsistent results but share a common shortcoming: inadequate attention to a key element in the persuasive process, namely, the personal relationship between the special-interest lobbyist and the government official—whether an elected official or an agency employee. We suspect that without considering that relationship, money spent on lobbying is more likely to be “wasted” in that it fails to produce the desired policy outcome. As Ansolabehere et al. (2003) suggest, because legislators are time-constrained, money's greatest utility for influencing policy is in buying access rather than securing a policy itself. Consequently, campaign contributions become crucial for securing appointments with officials (Austen-Smith, 1998). Tripathi et al. (2002) report evidence supporting Austen-Smith's claim, showing that special-interest groups that donate large amounts to political campaigns also emphasize lobbying and spend about ten times more on lobbying than on PAC contributions (Wright, 1990). While access does not guarantee policy change, it offers opportunities for informing legislators about policy consequences and building relationships. Therefore, money spent on gaining access is seen as the initial step toward developing a connection that may prove effective in influencing future policy.⁵

This last suggests a possible answer to the “Tullock Paradox,” which asks, “Why so little money in politics?” (Ansolabehere et al., 2003; Tullock, 1980). If a policy was decided purely in favor of the highest bidder – or if legislators posted prices for their votes (Bombardini & Trebbi, 2020), we would expect to observe more money being spent. That conclusion is particularly true if one takes the perspective of return on investment because studies such as Wagner and Elder (2021), Payson (2020), and Cox (2023) find incredibly large returns to campaign contributions. One limitation of those studies, however, is that *current* contributions are linked to legislative benefits without regard for the effort and dollars needed to build and maintain durable relationships. It would be interesting to re-examine the return on contributions angle using a humanomics perspective that accounts for current and past contributions as part of the investment required to earn a policy return. Said differently, donors who truly wish to influence policy may view the costs of attention and continued donations needed to forge a meaningful relationship to be lower than the expected benefits of their preferred policy.

At this juncture, it is important to define the term “relationship.” Going forward, we use the word to mean something based on more than either superficial aspects, such as fleeting emotions or financial considerations. Rather, a relationship comprises lasting and meaningful interpersonal utility, and developing one requires effort, commitment, and a willingness to invest in the connection over time.

1.1 Relationship contributions hypothesis

When special interests want to successfully influence public policy, they will buy access repeatedly over multiple years through contributions.

⁵ Alternatively, individuals could have consumption motives for donating grounded in shared ideologies (Ansolabehere et al., 2003). For example, no one would reasonably expect an individual contribution of \$50 to a presidential candidate to buy influence over a policy process or outcome.

Prior research by both political scientists and economists shows the economic importance of relationships to lobbying. That evidence makes the subject fruitful for investigation from the perspective of humanomics: the study of how human beings interact in both the personal and the transactional realms.

To enrich our insights, we build upon McCloskey's (2010, 2016a, 2016b; McCloskey & Silvestri, 2021) work on talking as an important part of relationship building.

Why would interest groups lobby people who may eventually become governor unless they want to build a relationship with the future governor in advance? If relationships are not important, it would be easier for lobbying groups to save their money until a new governor has been elected.

Evidence for the key role of relationships comes from the impact of the so-called revolving door in politics. For example, Vidal et al. (2012) find that one-time congressional-staffers-turned-lobbyists experience income losses when one of their contacts retires, dies, or loses reelection. The implication is that about 23% of a lobbyist's earnings come from relationships. The loss of earnings is consistent with a reduction in influence, making intuitive sense because such a pair—the lobbyist and the lobbied—will have developed a unique, mutually trusting relationship over time. Strickland (2023) finds that former Congress members turned lobbyists receive more income and achieve more favorable policy outcomes than other lobbyists. The premium those lobbyists earn is, however, contingent on the continued presence of former colleagues in legislatures.

In the same vein, legislators often raise funds through their own personal relationships, families, and networks of friends (Francia et al., 2003). Following Weingast et al. (1981), we think of politicians as being motivated by some combination of seeking reelection, seeking election to higher office, ideology, avarice, and/or interpersonal relationships. Those notions are closely aligned with the principal human motivators delineated by humanomics.

Furthermore, our consideration of the relative limitations of the usefulness of money versus relationships as influencers of policy is broadly in keeping with Smith (1759/1976) and Smith and Wilson's (2019) conviction that human motivation incorporates more dimensions (including beneficence, approbation, sympathy, guilt, and propriety) than mere wealth maximization.⁶

Indeed, many of today's economists fail to recognize the complexities of maximizing utility and instead seek to measure utility simply by, for example, cold hard cash or election victories. One example of the limitation is research that looks at the net present value of working in an industry job after having worked in government as a regulator of that industry, such as Makkai and Braithwaite (1992), Katic and Kim, (2024), and Chalmers et al. (2022). While a "revolving door" certainly is observed in many regulated industries, the present value approach reduces utility maximization to a dollar figure without allowing for many other considerations that affect decision-making around current and future employment. In the context of political lobbying, we see something similar, namely, that income, campaign contributions, and career opportunities are all important motivators for both potential influencers and the potentially influenced.

⁶ Our understanding of feelings of approbation, sympathy, and guilt comes from Adam Smith's *Theory of Moral Sentiments* (TMS), not from *The Wealth of Nations* (WN). We think of the TMS as the "handbook" for the personal realm and the WN as the handbook for the impersonal extended market order. Our thinking is based on the work of Terjesen and Willis (2016).

The motivating power of relationships also figures into legislators' memberships in political parties, which is important for several reasons. First, for example, members want approbation from fellow legislators. Next, the path to important committees and positions, as well as some types of funding, runs through the party. Finally, party members have multiple aims, but one of them is the strength of the party in full. One difficulty in measurement is that legislators have multiple competing relationships as well as multiple personal objectives to maximize.

Empirically, evidence at the state level suggests that party consolidation impacts policy, including per capita spending (Reed, 2006). Caplan (2001) also finds that every party tends to make government larger as its likelihood of electoral victory increases. In contrast, Campbell and Mitchell (2011) argue that as parties consolidate power by building larger majorities, they are less likely to maximize state budgets. The evidence suggests that larger parties are either more efficient at providing the services that voters want at low prices or that narrow-majority parties participate in more logrolling, which requires greater government spending (Campbell et al., 2007).

To assess the significance of interpersonal relationships and lobbying efforts empirically, we take advantage of the 50-state US laboratory and investigate campaign contributions over time, relying on a rich, transaction-level dataset from 2000 to 2015, which identifies individual donors and recipients. Hypothesizing that state legislative term limits will shift relative political power away from the legislative branch to the executive branch, we estimate a difference-in-differences model comparing repeat donor contributions to likely gubernatorial candidates' pre- and post-term (legislative) limits.⁷

Our results show that once political power shifts away from the legislative branch, the number of contributions and dollars given by repeated donors directed to would-be gubernatorial candidates increase significantly relative to states that have never imposed term limits. If the goal of donors is to influence policy, then our estimates are consistent with the notion that money is being spent as an inroad to relationship development, implying that a humanomics lens can complement traditional models of public choice and political economy.

2 Humanomics background

Our goal in this section is to present an argument consistent with humanomics as to why a relationship beyond something purely transactional exists between the legislator and lobbyist. In short, repeated contributions create opportunities for recurrent interactions, interactions on which a relationship and trust can build. Furthermore, such relationship-building might explain why an elected official could care about the wants of specific lobbyists and seek their approbation. It is to that line of argument we now turn our attention.

⁷ Voters may have implemented term limits to quell related issues such as overly ambitious legislators, careerism, pork-barrel spending, and excessive regulation. However, studies by Fiorina and Fiorina (1989) and Lopez (2003) find no evidence that term limits mitigate other inefficiencies in the legislative process. In a contrasting view, Erler (2007) reports evidence that spending levels increase in states after term limits are enacted. Relying on congressional voting behavior and a Supreme Court ruling as a natural experiment, Lopez (2002) shows that politicians' voting behavior is very different with and without term limits in place.

2.1 Relationships and trust

Traditional neoclassical Max-U economics does not consider the importance of human relationships or why and how they matter, but the humanomics approach does. We opt for the humanomics approach. In simplest psychological terms, it is well established that humans care more about the feelings of people they know than the feelings of strangers or abstractions (Smith & Wilson, 2019: p. 73) and that repeated play builds trust (Kurzban et al., 2001; Smith & Wilson, 2019). In the context of American political lobbying, we posit that “repeated play” amounts to repeated conversations and interactions between the lobbyist for the regulated, the regulator, the legislator, or all three parties.

Successful lobbying depends on mutual respect and benevolence, actual friendship, or the next best thing to it (Lowery, 2013). The idea of building relationships and trust through “social lobbying,” in other words lobbying contacts that take place outside the context of work or the office, is neither new nor uncommon (Grose et al., 2022). According to a field experiment run by Grose et al. (2022), legislators who were socially lobbied more frequently were more likely to express public support for the interest group’s preferred policy than were legislators lobbied only in their offices or without personal contact from a lobbyist. Relatedly, studies also suggest that political elites are more easily influenced when lobbying is conducted in places not easily observed by the public. When, for example, a meeting is held in a friendly environment where participants feel comfortable, more “informal talk” ensues (Walsh & Cramer, 2004); with informal conversations, the lobbyist and legislator build reciprocal trust (Grose et al., 2022).

Repeated lobbying leads to a relationship that goes beyond the professional, a relationship that becomes a social connection (Banfield, 1961). Generally, sharing social cues leads to more cooperation, both in experiments and in real life (Rand et al., 2015). For instance, McGarrity and Sutter (2000) found that money from more than one election cycle has more influence on roll call votes, which suggests that PAC expenditures are intended to have a durable effect rather than being simple spot-market or one-period prepayment contracts. Studying the National Rifle Association, McGarrity and Sutter (2000) found that “buying votes” required contributions over at least three election cycles. Kroszner and Stratmann (2005) explored repeat contributions from the legislators’ point of view and conclude that legislators with consistent policy agendas are more likely to attract repeat contributions.

The alliances implied by repeated contributions over time support the notion that relationships play a critical role in shaping political and policy outcomes. If, as Grose et al. (2022) find, lobbying leads to long-term relationships and friendships, those friendships may promote the public good in the sense that they “become a check on opportunistic behavior” (Boehm, 2007). Regulators and the regulated also depend on one another for salient information about complex issues that, in some situations, may serve the broader interests of the public (Rex, 2020). Such “cognitive capture” likewise may be present in the lobbyist-lobbied relationship, providing another mechanism through which these connections could potentially benefit the public (Rilinger, 2023).

It may also be useful to consider Boehm’s work on political spending in the context of Akerlof’s (1982) findings on gift-giving and exchange in labor-management relations. In Akerlof’s model, workers’ effort hinges on embedded norms about a fair day’s work such that firms paying above market-clearing wages benefit from more worker effort. While Akerlof was discussing the microeconomic foundations of macroeconomic relationships between employees and employers, the mutual sense of obligation between

the lobbyist and the government official would seem to be a good parallel, as would McCloskey's (2010, 2016a) consideration of the importance of "sweet talk" and persuasion in negotiations and sales. That "faculty of speech," as Adam Smith calls it, results in greater trust between the people involved. Thus, constant talking constantly builds trust. In a more general sense, the "rules of the game," as Greif (2006) points out, are informal and require continual renegotiation, another way of saying repeated work on the relationship. Grief's model and its attendant virtues also apply to the lobbyist and the lobbied.

2.2 Human bystanders

In *The Theory of Moral Sentiments*, Adam Smith wrote that, "What so great happiness as to be beloved, and to know that we deserve to be beloved? What so great misery as to be hated, and to know that we deserve to be hated?" Roberts (2015) summarizes Smith's philosophy as follows: "We want the love of others to be authentic in the sense that we don't want it on the cheap." In other words, we want to feel like the admiration and love of others is authentic. At the same time, Smith notes that because it is easy to fool oneself or rationalize behavior, we must avoid the temptation of succumbing to strategic flattery—a reminder that it is hard to be objective when you have a horse in the race.

Applying that insight to the lobbyist-lobbied relationship, we note that onlookers, in this case, voters, provide some value to the process when they grant or withhold approbation. Smith and Wilson (2019) examine the phenomenon in Chapter 3 when they consider "human bystanders." We normally think of legislators and lobbyists hiding their actions from rationally ignorant voters, the implication being that the actions are concealed to avoid disapprobation. But, when lobbying is out in the open, the insinuation is that the uneducated bystander, the voter, can be convinced to approve. Hence, we see the formation of strange coalitions and even industries lobbying in favor of their own regulation (Horpe-dahl, 2021).

Certain types of lobbying might receive approbation from both the public at large and the regulated because the lobbying appears to be in the public interest. The implication is that humans—up to and including lobbyists and the interests they represent—see themselves as good or not, in part depending on how others see them. As an example of that dynamic, double anonymity reduces offers in the ultimatum game (Hoffman et al., 1994). In other words, people behave differently when the experimenter can observe their actions, suggesting that people want to be good and to appear to be good ("praiseworthy" in Adam Smith's terms). In the well-known article that lays out the Tullock paradox, Tullock (1980) emphasizes that firms do not want to appear avaricious, that they want a "cover story" for the general public. In fact, the need for a cover story explains why Baptists are so important in the Baptists and bootleggers theory (Yandle, 2022).

Adam Smith's most famous quotation probably is: "It is not from the benevolence of the butcher, the brewer, or the baker, that we expect our dinner, but from their regard to their own interest. We address ourselves not to their humanity but to their self-love, and never talk to them of our own necessities, but of their advantages." However, it is important to note that Smith elsewhere emphasizes that humans pursue not only their own self-interest, but that they also have sympathy:

How selfish soever man may be supposed, there are evidently some principles in his nature, which interest him in the fortune of others, and render their happiness nec-

essary to him, though he derives nothing from it except the pleasure of seeing it.
(Smith 1789/1976)

TMS notes that some people's approbation matters more than others, and not necessarily because those people are especially honorable. In the case of political lobbying, the implication may be that over an unspecified time horizon—maybe a short one—the lobbyist and the lobbied actually come to admire one another and thus to value one another's approbation. Smith notes further, unhappily, that humans often admire the rich and powerful. Applying that notion to the lobbying process suggests that lobbyists wear expensive suits and ties because appearing wealthy makes approbation and a relationship built on it more likely.

The disposition to admire, and almost to worship, the rich and the powerful, and to despise, or, at least, to neglect persons of poor and mean condition, while necessary both to establish and to maintain the distinction of ranks and the order of society, is, at the same time, the great and most universal cause of the corruption of our moral sentiments (Smith, 1759/1976).

3 Data & empirical strategy

Having established the humanomics argument for why a personal relationship between the legislator and lobbyist is likely to emerge and be valued, we now turn our attention to providing empirical support for the existence of the claimed relationship. Measuring the strength of the relationship and testing the impact of how varying degrees of relationship strength might impact outcomes, while intriguing and still needed, is beyond the scope of this paper.⁸

3.1 Data

To explore our relationship-contributions hypothesis empirically, we leverage data from the National Institute for Money in Politics, hereafter NIMP. NIMP compiles and cleans individual campaign contributions from campaign finance reports.⁹ They graciously provided us with records of individual contributions from 2000 to 2015 for every US state. The records include the recipient candidate or organization, office sought (if applicable), whether the recipient is an incumbent or challenger (if applicable), the dollar value of the contribution, the election jurisdiction, the address and business line of the contributor, and whether the donor resides in-state or out-of-state. Recipients and contributors, whether they are individuals or organizations, also are assigned unique identifiers, making it possible to track their behavior over time.

⁸ Further extensions of this line of research include examining how personal relationships and their strengths impact the efficiency of public policy, citizen well-being, the size of government, and total lobbying expenditure. Indeed, those extensions could serve as the foundation of a rewarding research agenda.

⁹ The National Institute for Money in Politics provides free access to some of their aggregated data on state and federal contributions at: www.followthemoney.org.

The NIMP data have been utilized in many academic studies on topics ranging from teacher unions to judicial decisions and from corporate tax rates to state telecommunications regulatory policies (see, for example, Wagner & Elder., 2021; Chirinko & Wilson, 2010; Fourniaies & Hall, 2014).¹⁰ However, to our knowledge no prior study has leveraged NIMP's dataset to study the repeat-contribution behavior of individuals or organizations.

Unlike previous studies (at the federal level) on repeat contribution behavior, we are the first to our knowledge to rely on quasi-experimental methods. We capitalize on state-level variation in legislative term limits that we (and others) argue will shift the relative power between different branches of government. Our quasi-natural experiment focuses on how repeat-contribution behavior changes in states after they impose terms relative to states whose legislative structure remains unaltered.

Prior to term limits, we see a situation wherein senior legislators have tremendous power through acquired institutional knowledge and agenda control (Cox & McCubbins, 2005). In other words, senior legislators serve as repositories of complex and sticky knowledge that they sometimes exploit to their own ends.¹¹ While governors often set the overall agenda for a state, legislators may or may not play ball. Senior legislators can pull many powerful levers, even if, as Alexander Hamilton wrote in *Federalist* 73, the veto is a “shield to the Executive.” Once senior legislators cease to be quite so senior – having been term-limited out of office, for example—the governor is working with and competing against weaker foes, giving him or her a relative advantage. Thus, term limits redistribute power away from senior leaders in the legislature and toward governors (Carey et al., 1998). Said differently, we (and others) argue that legislative term limits will shift the equilibrium balance of political power between branches.¹² For this reason, we argue that term limits provide a viable path for gauging the importance of relationships in political settings. As evidence of their relative importance, studies such as Hojnacki and Kimball (1998), Hojnacki and Kimball (2001), and Lorenz (2020) have demonstrated that more powerful legislators attract greater attention and contributions from lobbyists than their less-influential peers.

As noted previously, the national institute for money in politics (NIMP) relies on information submitted in campaign finance reports to identify distinct donors. Each donor is assigned a unique identification code, allowing us to track their contributions over time. After removing self-contributions from candidates, we construct two distinct measures of repeat contributions: (a) the fraction of all unique donors in year t who donated in a prior year, and (b) the fraction of all dollars received in year t from donors who contributed in a prior year. Since our objective is gauging repeat donations to key or potential power brokers in the executive branch (i.e., individuals with higher probabilities of remaining or becoming governor), we aggregate donations (across primary and general elections) to all candidates in all races for governor, lieutenant governor, attorney general, and secretary of state.

¹⁰ The National Institute for Money in Politics commissioned the Rand Corporation to evaluate the data and document how they are being used. The very comprehensive report, authored by McGovern and Greenburg (2014), is available at: https://www.rand.org/content/dam/rand/pubs/research_reports/RR700/RR791/RAND_RR791.pdf (accessed: 3/21/2023).

¹¹ We are using the term senior generically to refer to powerful or influential legislators.

¹² According to the National Conference of State Legislatures, 37 states impose a term limit on the governor. The last state to change its gubernatorial limits was Wyoming in 1992. Hence, regardless of whether a state does or does not impose gubernatorial term limits, the adoption of legislative term limits could disrupt the equilibrium balance of political power, all else equal.

Since the contribution data do not begin until 2000, the earliest date for which we could construct the repeat donation measures is 2001. In each subsequent year, we sequentially expand the list of past donors to reflect the full history of contribution activity. For example, when repeated donor counts and dollars are constructed for, say, 2007, every prior donation from 2000 to 2006 is considered.

The treatment group is defined as all states that introduced legislative term limits during our sample period, with the treatment date being defined as the first year those term limits became binding in any legislative chamber. The comparison group consists of states that have never enacted any form of legislative term limits. Ten states imposed term limits that became binding before our dataset begins (pre-2000), so they must be excluded from the analysis. The treated, comparison, and excluded states are shown in Fig. 1.

For the treated and comparison groups in our sample, Fig. 2 shows the fraction of donors and dollars from repeat donors annually between 2001 and 2015. Over the full sample, repeat donors, both in terms of counts and dollars, are similar across treated and comparison states. The data show that, for instance, repeat donors accounted for an average of 17.1% of all donors in the comparison states and about 30% of dollars donated. In treated states, the comparable figures are 22.0% and 33.5%, respectively. For three years in the sample, 2009, 2010, and 2011, the fraction of total dollars contributed in treated states exceeded 50%. Those statistics suggest a steady influx of new donors across state election cycles.

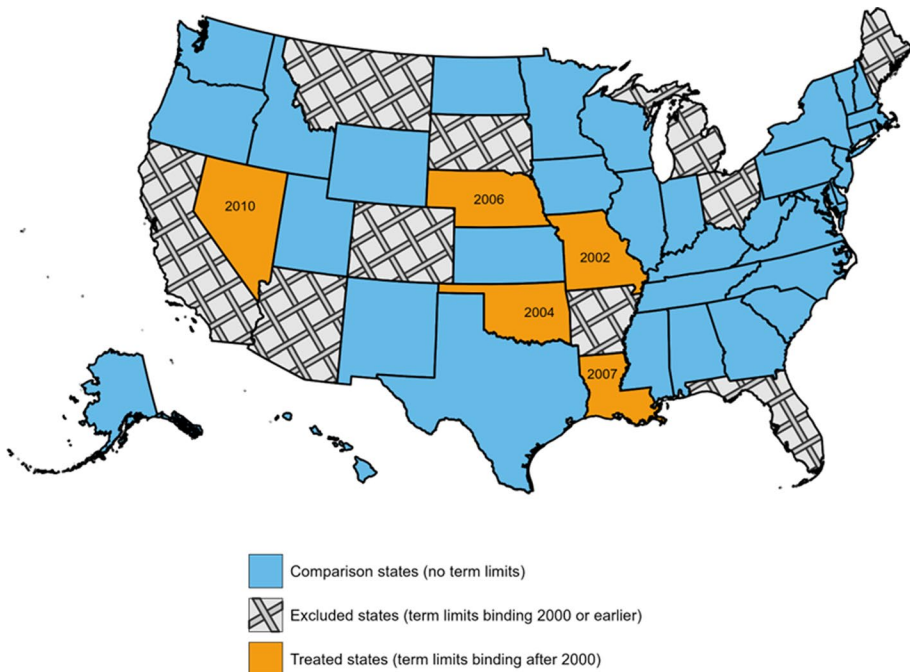


Figure1 States with and without binding legislative term limits. Source: National Conference of State Legislatures. The year shown for selected states is the first year that state legislative term limits became binding. We must exclude states for which term limits became binding before 2001 from our empirical analysis because we lack pre-treatment data for these states. See Sect. 3.1 for more details

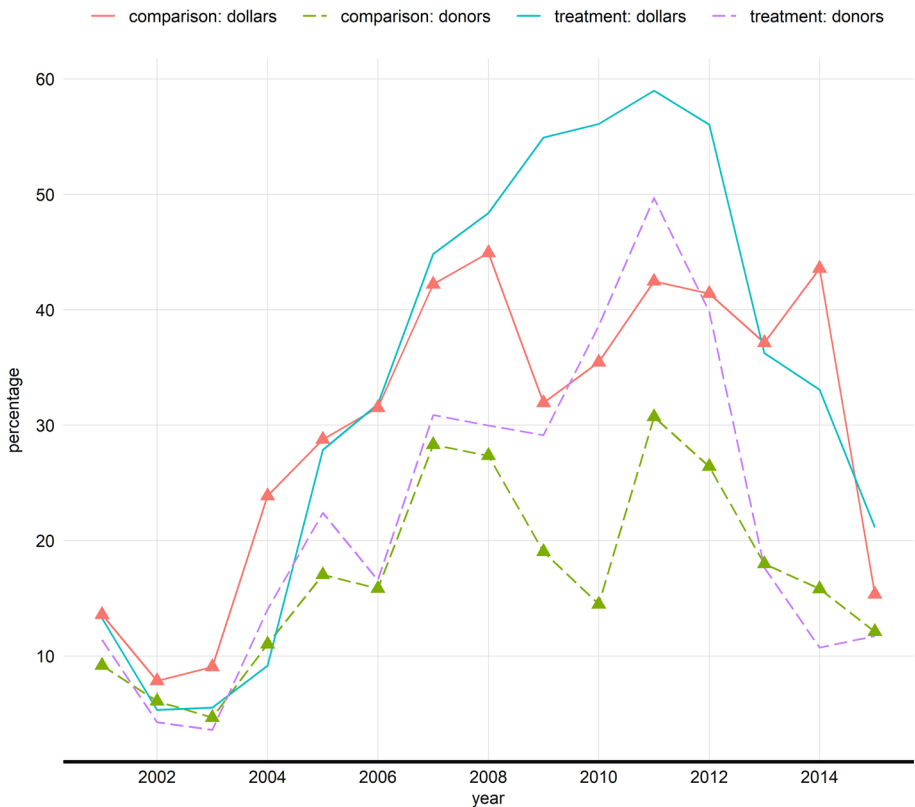


Fig. 2 Campaign contribution: donors and dollars from repeat contributors. Source: Authors' calculations using data from the National Institute for Money in Politics. Dashed lines show the fraction of total distinct donors (in given years) to candidates for governor, lieutenant governor, attorney general, or secretary of state who are repeat donors (since 2000). Solid lines show the fraction of total dollars donated (in given years) to candidates for governor, lieutenant governor, attorney general, or secretary of state who are repeat donors (since 2000). Comparison group state lines are those with the triangle markers. See Fig. 1 for the states in the treatment and comparison groups

3.2 Empirical specification

Considering that we have a sample of states with and without binding legislative term limits in place, one approach to estimating the causal effect of the political importance of relationships is to pursue a difference-in-differences strategy. With panel data, that model can be expressed as:

$$R_{i,t} = \alpha + \gamma_i + \lambda_t + \delta D_{i,t} + \theta X_{i,t} + \epsilon_{i,t}, \quad (1)$$

where $R_{i,t}$ is the repeat contribution outcome of interest in state i at time t , γ_i are state fixed effects, λ_t are fixed year effects, and $D_{i,t}$ is an indicator variable that equals unity for states that have binding legislative term limits in the post-treatment periods (2001, or later). Because of data limitations, we must exclude states for which term limits became binding in 2000 or earlier.

As illustrated in Fig. 1, the timing of when term limits first become binding varies from state to state. As Goodman-Bacon (2021) has demonstrated, if treatment dynamics or heterogeneity materialize across treated states, then a conventional two-way fixed effects (TWFE) estimator may be biased in the presence of staggered adoptions because states treated earlier in the sample ultimately serve as part of the comparison group for states treated later in the sample. If the effects across states or over time are heterogeneous, then the TWFE estimator will not recover the average treatment effect on the treated (ATT).¹³

We follow Cengiz et al. (2019) and employ a “stacked regression” difference-in-differences approach to address the staggered adoption of legislative term limits. The approach is very straightforward to implement and, as Baker et al. (2022) show, yields an unbiased estimate of the average treatment effect even if heterogeneous treatment effects are present. While other estimators have been proposed recently to deal with staggered timing (Callaway & Sant’Anna, 2021; Sun & Sant’Anna, 2021), the stacked regression approach is numerically equivalent (see Baker et al., 2022), intuitive, and can be estimated by ordinary least squares (OLS).

The underlying idea of the stacked regression strategy is the formation of a comparison group for each treatment group cohort, so that “early” treated states never become part of the comparison group. That objective can be fulfilled simply by creating a “dataset” for each treatment group cohort based on the year term limits became binding, converting each cohort’s dataset into relative time, and then combining the cohort datasets.

Given that the observations on individual campaign contributions do not begin until 2000, we are limited to exploring five treated states (Louisiana, Missouri, Nebraska,¹⁴ Nevada, and Oklahoma) wherein term limits became binding between 2002 and 2010. Thirty-five states, shown in blue in Fig. 1, never have enacted any form of legislative term limits. Those states are “never treated” and form the base comparison group.¹⁵

To illustrate the formation of a cohort dataset, consider Nebraska, for which term limits became binding in 2006. That cohort’s dataset consists of all treated states wherein term limits became binding in 2006 (which, in this instance, is only Nebraska) and all 35 states that never enacted term limits.¹⁶ Treatment occurs at time 0 (or 2006) and relative times -1 and $+1$ refer to 2005 and 2007, respectively. The same process is repeated for the four other treated states and the final dataset is formed by “stacking” (row concatenating) the individual cohort datasets. The data structure accomplishes two objectives. First, it guarantees that no treated units are ever part of the comparison group. Second, it guarantees that every treated cohort is treated in the same year (period 0).¹⁷

¹³ Baker et al. (2022) ran simulations and showed that the two-way fixed-effects estimator is biased if the treatment effect is dynamic (changing over time) or heterogeneous across treated units. Under the strict condition that any treatment effect is constant over time and across space, the two-way fixed-effects estimators will recover the average treatment effect on the treated without any additional modifications. However, the assumption of constant average treatment effects across states and time is a stringent one.

¹⁴ Nebraska is uniquely unicameral.

¹⁵ Idaho, Massachusetts, Oregon, Utah, Washington, and Wyoming all enacted legislative term limits and later rescinded them before they became binding. As a precaution, we estimated our regression models excluding those states from the comparison group. Our results are robust to that exclusion.

¹⁶ In our application, every treated cohort consists of a single state because the year in which term limits became binding is different for every treated state. If, on the other hand, term limits became binding in two states in, say, 2006, then the 2006 cohort would consist of the two treated states and 40 “never treated” states.

¹⁷ One noteworthy issue arises with the stacked regression difference-in-differences approach. As Cengiz et al. (2019) and Baker et al. (2022) observe, the regression must be fully saturated with state- and year-fixed effects for each specific cohort dataset.

If the treatment date for any treated cohort, g , is re-defined as period 0, then cohort g 's dataset comprises treated *and* comparison observations from -5 to $+8$. If we are unable to include -5 to $+8$ because of the timing of binding term limits, we enter as many observations as possible within those lags and leads. Observations would be restricted, for instance, if we observe fewer than five pre-treatment periods or fewer than eight post-treatment periods. We adopt five pre-treatment periods because that window fully encompasses one full gubernatorial election cycle before term limits become binding. Similarly, we adopt eight post-treatment periods because that window encompasses two gubernatorial cycles after term limits become binding.

Following Wagner and Elder (2021), we control for several time-varying state covariates ($X_{i,t}$) that have been shown to be correlated with political campaign contributions. To control for differences in economic conditions and other policies, we enter per capita personal income (in thousands), the share of the state's population aged 65 and older, an indicator variable for the political party of the governor, an indicator variable for gubernatorial election years, and the state's citizen ideology score (Berry et al., 2013).¹⁸

To adjust for differences in electoral competitiveness, which could drive more contributions, we enter two variables. First, we control for Calcagno and Lopez's (2012) divided government score to capture the degree of political unification between the executive and legislative branches.¹⁹ Second, following Wagner and Elder (2021), we include a measure of electoral competitiveness in the state, defined as the median value of the inverse of the Herfindahl–Hirschman index (HHI) of vote shares for every legislative seat. If, for example, four candidates receive the same vote share for a given legislative seat (25% each), then the inverse HHI = 0.005. We enter the HHI's inverse so that larger values indicate more competitiveness in the median legislative seat in the state. This measure is also party-independent.

Finally, we include two variables to capture differences in state campaign finance systems. The first, *corporate contributions restricted*, equals unity if the state imposes any form of limit on direct political contributions from corporations. The second, *public financing of elections*, is an index reflecting the degree to which public funding can be accessed by individual candidates. The index ranges between 0 and 7, with higher values indicating more generous access to taxpayer-financed campaign funding.²⁰ Descriptive statistics and the sources of information for each variable are displayed in Table 1.

¹⁸ The Berry et al. (2013) ideology scores range from 0 to 100, with 100 representing the most liberal value.

¹⁹ The Calagno-Lopez divided government score (DG) is a continuous measure that is equal to $DG = -\log [(g_{it} + 0.5)/(n_{it} - g_{it} + 0.5)]$, where g_{it} is the number of legislators in state i and time t in the same party as the governor and n_{it} is the total number of legislative seats. If the legislative is evenly divided party-wise, then $DG = 0$.

²⁰ See the Correlates of State Policy database for greater detail. Each state's index (in year t) is given one point for each of the following rules/legislation: a total campaign expenditure limit; check-off on income tax returns for contribution to public financing; independent revenue source for public financing; public financing of statewide campaigns; public financing of state legislative campaigns; public financing of political parties; and equal distribution of public funds across candidates and parties.

Table 1 Descriptive statistics

Variable	Mean Treatment	Std. dev	Mean Comparison	Std. dev
Repeat donor share: count	22.03	24.83	17.06	20.34
Repeat donor share: in-state count	23.17	25.18	17.37	20.52
Repeat donor share: dollars	33.51	30.16	29.94	29.17
Repeat donor share: in-state dollars	34.46	30.54	30.98	30.09
Per capita personal income	36.66	5.82	39.29	8.62
Calagno-Lopez divided gov score	0.03	0.47	- 0.17	0.77
Citizen ideology score	38.71	11.35	53.20	16.88
Public financing of elections	0.29	0.71	1.52	1.75
Corporate contributions restricted	0.91	0.29	0.84	0.37
Gubernatorial election year (1 = yes)	0.25	0.44	0.27	0.44
Electoral competitiveness	6.21	10.35	8.33	13.67
Governor party (1 = Democrat)	0.35	0.48	0.63	0.48
Share of population 65 +	13.14	1.10	13.12	1.92

Descriptive statistics for treated and comparison states are from 2001 through 2015. Repeat donor share: count and dollars are the fraction of all distinct contributors or total dollars made in the state for all candidates for governor, lieutenant governor, attorney general, and secretary of state. In-state removes any donor (individual or organization) located outside of the state where the election jurisdiction is located. Population share 65 and older is from the Census Bureau. Per capita GDP (\$000 s) is from the Bureau of Economic Analysis. Citizen ideology scores are from Berry et al. (2013) and available at <https://rcfording.com/state-ideology-data/>. The Calagno and Lopez (2012) divided government score, gubernatorial election years, and governor party were constructed from data from various issues of The Book of the States. State electoral competitiveness measures are from Carl Klarner, State Legislative Election Returns, 1967–2016 available at the Harvard Dataverse <https://doi.org/10.7910/DVN/3WZFK9>. Finally, information on public financing of elections and whether the state restricts corporate campaign contributions are from the Correlates of State Policy database (<http://ippsr.msu.edu/public-policy/correlates-state-policy>)

4 Empirical results

Our baseline empirical estimates are reported in Table 2. The outcome of interest in columns 1 and 2 is the fraction of contributions from repeat donors. Column 1 includes all donors, while column 2 includes only those donors who reside in the same state as the recipient. Similarly, columns 3 and 4 measure repeat contributions as the fraction of contribution dollars coming from the same donors. Column 3 includes all donations, while column 4 limits donations to donors who reside in the same state as the recipient.²¹ A constant term and year-state by cohort fixed effects are entered in each regression but not reported.

The difference-in-differences parameter of interest, labeled as “D” in Table 2, measures the average difference in the outcome between the treated and comparison states before and after term limits become binding. Consistent with our hypothesis, we find that repeat contributions—both counts and dollars—are increasingly important for candidates who may be most likely to stay or become governor *after* legislative term limits become binding.

²¹ In general, a very large majority (90%+) of donors and dollars come from groups or individuals who reside in the same state where the election is taking place. Anecdotally, out-of-state contributions seem to be more prevalent in swing states during close gubernatorial elections. In all the empirical specifications, the electoral jurisdiction is statewide.

Table 2 Effect of binding term limits on repeat donations

	<i>Dependent variable</i>			
	Repeat donors	Repeat donors	Repeat donors	Repeat donors
	Count	In-state count	Dollars	In-state dollars
	(1)	(2)	(3)	(4)
	OLS	OLS	OLS	OLS
D	4.766 (5.927)	7.371 (5.169)	11.022*** (3.713)	11.558*** (3.934)
Per capita personal income	0.630 (0.409)	0.040 (0.409)	-0.071 (0.616)	-0.653 (0.640)
Calcagno-Lopez divided gov score	-2.012** (0.905)	-1.683* (0.977)	-3.049*** (1.130)	-2.654** (1.287)
Citizen ideology score	0.035 (0.074)	0.169** (0.082)	0.344*** (0.117)	0.409*** (0.126)
Public financing of elections	-5.715*** (1.088)	-5.799*** (1.232)	-7.079*** (1.975)	-7.377*** (2.124)
Corporate contributions restricted	-1.404 (1.386)	-2.001 (1.340)	-7.602*** (2.454)	-7.419*** (2.412)
Gubernatorial election year (1 = yes)	-8.505*** (1.038)	-8.150*** (1.123)	0.062 (1.602)	0.887 (1.593)
Electoral competitiveness	0.087** (0.043)	0.100** (0.042)	0.229*** (0.058)	0.226*** (0.054)
Governor party (1 = Democrat)	2.157 (1.422)	2.386 (1.552)	6.751*** (2.081)	7.280*** (2.325)
Share of population 65 +	2.930 (2.309)	0.792 (2.127)	-2.841 (2.978)	-2.554 (2.953)
N	2,196	2,196	2,196	2,196
Year FE	yes	yes	Yes	yes
State FE	yes	yes	Yes	yes
In state only	no	yes	No	yes

This table presents the effect of binding legislative term limits on the fraction of donors and dollars to candidates for governor, lieutenant governor, secretary of state, and attorney general that come from repeat contributors. D denotes the difference-in-difference coefficient of interest. The regressions rely on the stacked difference-in-differences approach proposed by Cengiz et al. (2019). Each regression includes a constant term, year-by-cohort fixed effects, and state-by-cohort fixed effects that are not reported. Standard errors are in parentheses and are clustered at the state dimension. *** denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level

For contribution counts, shown in columns 1 and 2, the point estimates indicate that the fraction of repeat donors increases by 4.8 percentage points among all donors and 7.3 percentage points among in-state donors. However, the point estimates are too imprecisely measured to reject the null hypothesis that they differ from zero at conventional significance levels.

In terms of contribution dollars, shown in columns 3 and 4, the estimates suggest that the fraction of donations from repeat donors increases between 11.0 and 11.6 percentage

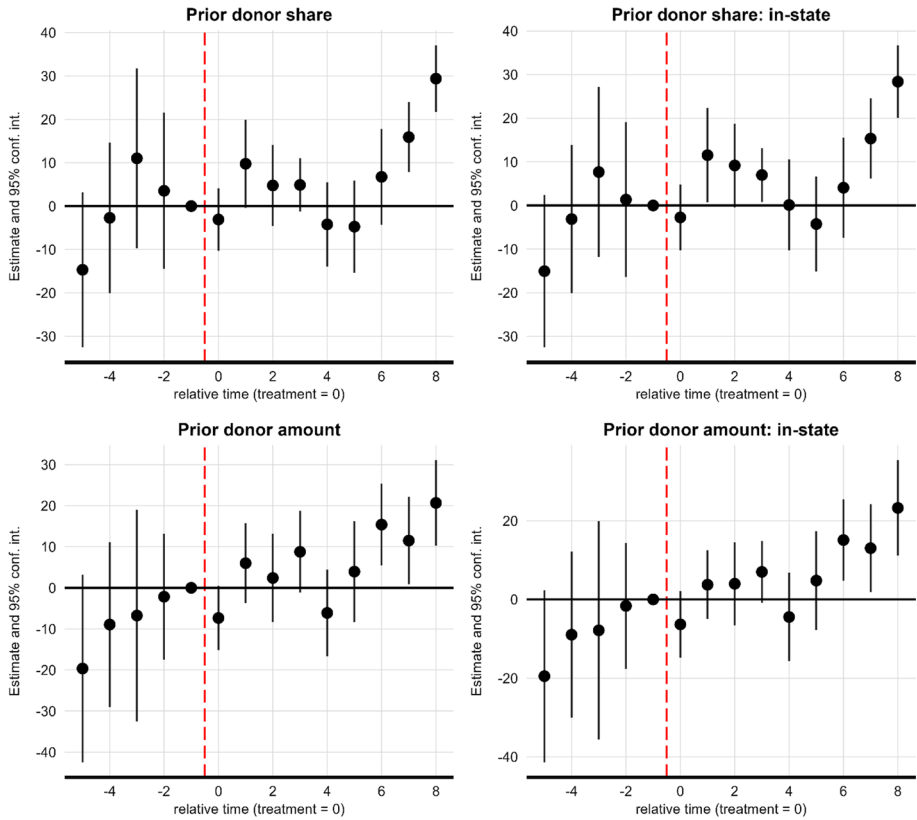


Fig. 3 Event study regression formulations. Three figures show the event study formulations of the regression results in Table 2. Error bars indicate 95% confidence intervals. The period just before treatment, -1 , is the omitted time indicator

points *after* term limits are binding relative to states that never have imposed term limits. Both point estimates are statistically significant at the one percent level. Additionally, considering that—on average—one-third of contribution dollars come from repeat donors in the full sample, the point estimates suggest that dollars from repeat donors increase by approximately 33% when political power shifts from the legislative to the executive branch. That is an economically meaningful effect and provides evidence consistent with our conjecture that relationships play a central role in molding the modern political process.

The final aspect to consider in the empirical specifications is the parallel-trends assumption. To assert a causal claim from a difference-in-differences design, one must be willing to assume that, in our setting, the behavior of repeated donations after term limits in the comparison states is a suitable counterfactual for how donations would have evolved in the treated states had term limits never been imposed. The counterfactual is impossible to test, of course, because we never get to observe how donations *would have evolved* in the states that enacted term limits had they failed to do so.

We follow best practices and estimate event study formulations of the difference-in-differences to probe the validity of the parallel-trends assumption holding in the pre-treatment period. If no evidence emerges of a difference in repeat donation behavior between the

treated and comparison states pre-treatment (before term limits became binding), then one may have some confidence that the (untestable) parallel-trends assumption may be valid.

Figure 3 shows the event study plots, with error bars indicating 95% confidence intervals. In each specification, the year before treatment is the omitted indicator variable. We find no evidence of any significant pre-treatment differences in repeated contribution behavior between the treated and comparison states. Thus, once political power shifts between branches, we feel confident that we have identified a causal estimate of the importance of personal relationships in political settings.

4.1 Limitations

Our analysis does not rely on enacted policies or roll-call votes and thus does not shed light on the important question of whether repeated lobbying influences policy outcomes. Rather, the present study focuses exclusively on the behavior of special interest lobbyists who are, it is said, “among the most experienced, knowledgeable, and strategic actors one can find in the everyday practice of politics” (Hall & Deardorff, 2006).

Likewise, with data limited to financial metrics, we have no way to measure the strength of the relationship between lobbyists and the lobbied. We do not know, for example, if a particular lobbyist has ever been invited to a particular state official’s family wedding. We do note, however, that Francia et al. (2003) describe the large number of New Jersey politicians who attended the circumcision of a donor’s grandson as one anecdote.

5 Conclusion

We began by wondering about the role of special interest money when it comes to influencing legislative outcomes. Some research finds that it matters; some research finds that it does not. Further research asks why, given the stakes, so little money is evident in politics in the first place (Ansolabehere et al., 2003; Tullock, 1980). To deepen our understanding of what money does and how it does so, we adopted a novel approach—combining existing public choice and political economy theories with elements of humanomics from Adam Smith, Vernon Smith, and Deidre McCloskey—to consider the likelihood that special interest lobbyists spend money as inroads toward developing relationships with officials in state government. To be successful, lobbyists must build the relationship before the next election. In our study, we looked at the lobbying of people likely to stay in the governor’s mansion or become state chief executives in the future.

Viewed through the humanomics lens, the relationship would involve the official seeking the approbation of the lobbyist, empathizing with the lobbyist (Rilinger, 2023), and perhaps feeling beneficence towards the lobbyist. As a means of testing special-interest lobbyists’ relationship-forming behavior, we asked whether the number and amount of repeat campaign contributions shifted away from legislators to members and potential members of the state executive branch when those officials became more important politically owing to the imposition of legislative term limits. We find that, indeed, the focus does shift.

This study implies further that lobbying may be more than mere exchange (vote buying or the promise of a future job), persuasion, informative signaling, or the sharing of information and political intelligence. Rather, lobbying is a mix of all those things, with the goal of forming alliances to achieve special-interest policy goals.

In sum, the study at hand supports the notion that money buys access, that access over time builds relationships, and perhaps that relationships influence policy. Beyond the political realm, we believe that our work deepens our understanding of the broad scope of human interactions, with political lobbying being just one example.

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